PVsyst - Simulation report

Standalone system

Project: WAJIR

Variant: New simulation variant
Standalone system with batteries
System power: 81.9 kWp
Wajir - Kenya

Author

PVsyst V8.0.6
VC0, Simulation date:
31/08/26 16:27
with V8.0.6

Project summary

Geographical Site
Wajir
Kenya

Situation

Latitude	1.75 °N
Longitude	40.02 °E
Altitude	252 m
Time zone	UTC+3

Project settings

Albedo	0.20

Weather data
Wajir
Meteonorm 8.2 (1991-2007), Sat=100% - Synthetic

System summary

Standalone system

Standalone system with batteries

Orientation #1
Fixed plane
Tilt/Azimuth 20 / 0 °

Orientation #2
Fixed plane
Tilt/Azimuth 20 / 0 °

User's needs
Daily household consumers
Constant over the year
Average 331 kWh/Day

System information

PV Array

Nb. of modules	140 units
Pnom total	81.9 kWp

Battery pack

Technology	Lithium-ion, LFP
Nb. of units	6 units
Voltage	614 V
Capacity	600 Ah

Results summary

Useful energy from solar	117.00 MWh/year	Specific production	1429 kWh/kWp/year	Perf. Ratio PR	64.51 %
Missing Energy	3.82 MWh/year	Available solar energy	145.54 MWh/year	Solar Fraction SF	96.84 %
Excess (unused)	25.02 MWh/year				

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General parameters

Standalone system

Standalone system with batteries

Orientation #1
Fixed plane
Tilt/Azimuth 20 / 0 °

Sheds configuration
No 3D scene defined

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Orientation #2
Fixed plane
Tilt/Azimuth 20 / 0 °

Sheds configuration
No 3D scene defined

User's needs
Daily household consumers
Constant over the year
Average 331 kWh/Day

PV Array Characteristics

PV module
Manufacturer Jinkosolar
Model JKM585M-7RL4-V
(Original PVsyst database)
Unit Nom. Power 585 Wp
Number of PV modules 140 units
Nominal (STC) 81.9 kWp
Modules 7 string x 20 In series
At operating cond. (50°C)
Pmpp 74.7 kWp
U mpp 806 V
I mpp 93 A

Controller
Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.
Converter
Maxi and EURO efficiencies 97.0 / 95.0 %
Battery Management control
Threshold commands as SOC calculation
Charging SOC = 0.96 / 0.80
Discharging SOC = 0.10 / 0.35

Battery
Manufacturer Deye
Model BOS-G 12 - 61.44 kWh
Technology Lithium-ion, LFP
Nb. of units 6 in parallel
Discharging min. SOC 10.0 %
Stored energy 331.8 kWh
Battery Pack Characteristics
Voltage 614 V
Nominal Capacity 600 Ah (C10)
Temperature Fixed 20 °C

Total PV power
Nominal (STC) 82 kWp
Total 140 modules
Module area 383 m²

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 144 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss
Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

Module Quality Loss
Loss Fraction -0.8 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

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Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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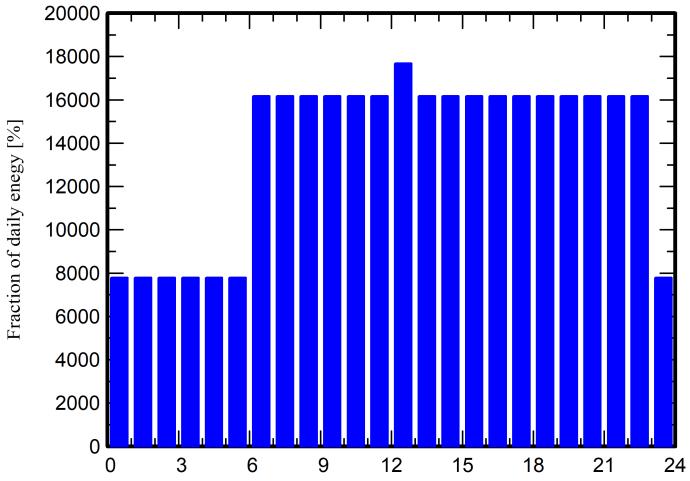
Detailed User's needs

Daily household consumers, Constant over the year, average = 331 kWh/day

Annual values

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	57	100/lamp	17.0	96900
Cashier's Computers	4	150/app	17.0	10200
Drink Fridges	6	870/app	24.0	125280
Fridge / Deep-freeze	3		24	41040
Wall Mount Fans	35		17	35700
Security Lights+Signage+8 channel	1	IP 850 tot	24.0	20400
Water Pump	1	1500 tot	1.0	1500
Total daily energy				331020

Hourly distribution



Main results

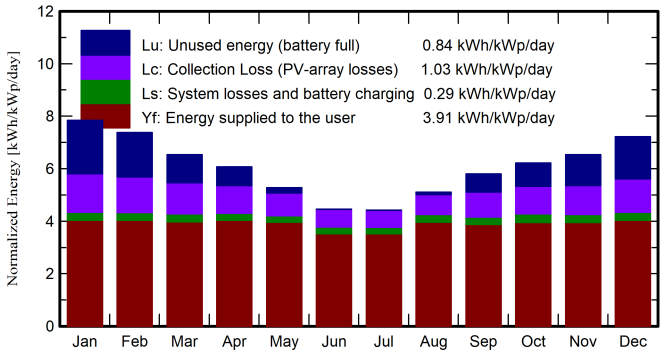
System Production

Useful energy from solar	117.00 MWh/year	Perf. Ratio PR	64.51 %
Available solar energy	145.54 MWh/year	Solar Fraction SF	96.84 %
Excess (unused)	25.02 MWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	3.6 %	Cycles SOW	96.1 %
Missing Energy	3.82 MWh/year	Static SOW	90.0 %

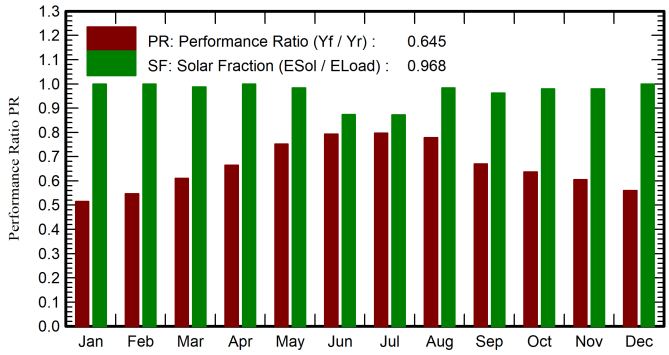
Economic evaluation

Investment		**Yearly cost**		**LCOE**	
Global	2,641,760.00 Ksh	Annuities	0.00 Ksh/yr	Energy cost	0.00 Ksh/kWh
Specific	32.3 Ksh/Wp	Run. costs	0.00 Ksh/yr		
		Payback period	Unprofitable		

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

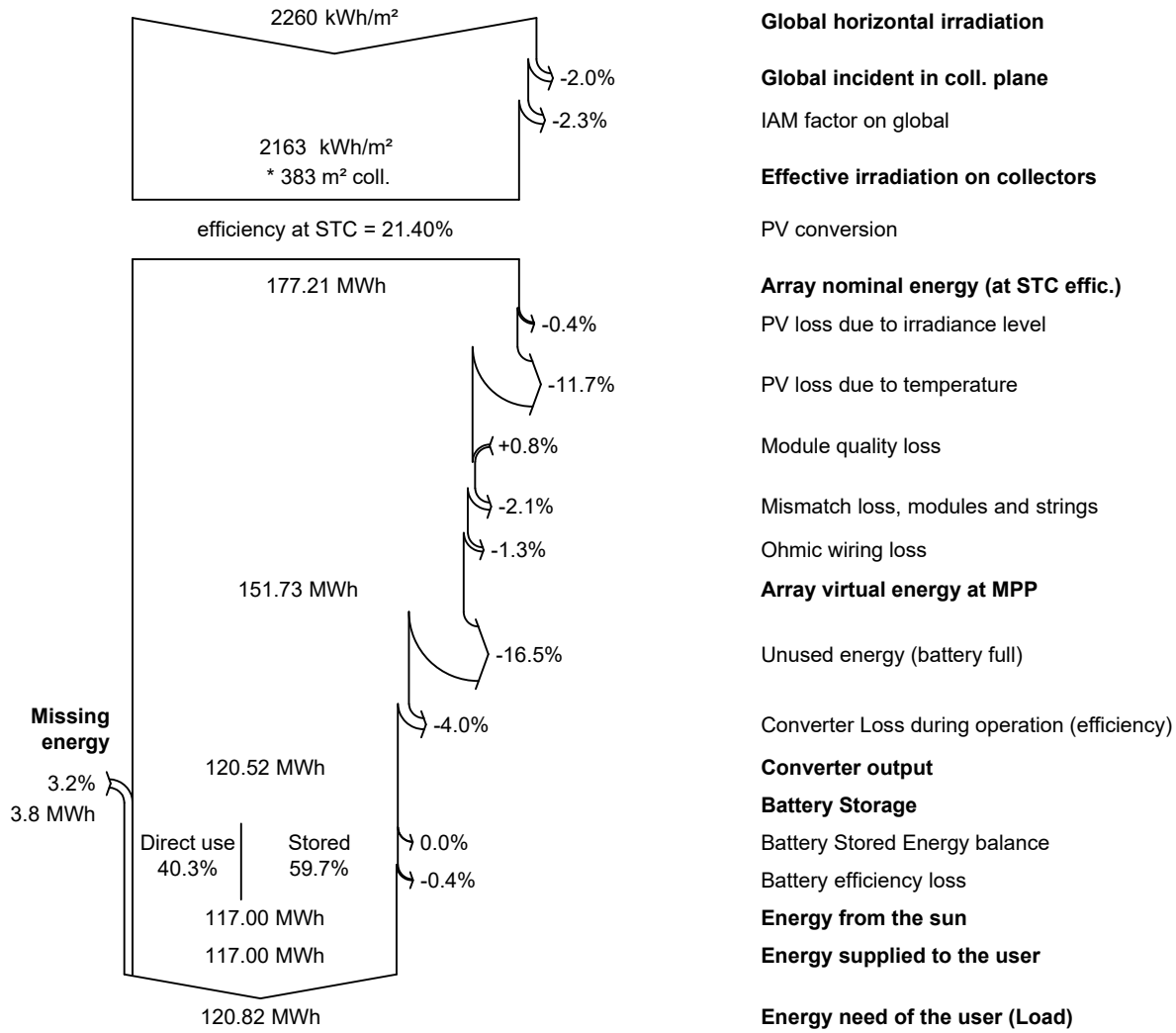
	GlobHor kWh/m²	GlobEff kWh/m²	E_Avail MWh	EUnused MWh	E_Miss MWh	E_User MWh	E_Load MWh	SolFrac ratio
January	215.0	239.6	15.77	5.15	0.000	10.26	10.26	1.000
February	194.2	203.5	13.42	3.86	0.000	9.27	9.27	1.000
March	206.7	198.2	13.17	2.70	0.130	10.13	10.26	0.987
April	202.3	177.4	11.94	1.75	0.000	9.93	9.93	1.000
May	194.8	158.1	10.78	0.51	0.175	10.09	10.26	0.983
June	161.8	128.7	8.92	0.00	1.249	8.68	9.93	0.874
July	162.0	132.2	9.21	0.00	1.308	8.95	10.26	0.873
August	178.5	153.4	10.60	0.21	0.175	10.09	10.26	0.983
September	182.8	169.9	11.49	1.67	0.376	9.55	9.93	0.962
October	187.3	189.0	12.66	2.21	0.208	10.05	10.26	0.980
November	179.1	192.8	12.92	2.90	0.202	9.73	9.93	0.980
December	195.7	220.4	14.66	4.05	0.000	10.26	10.26	1.000
Year	2260.0	2163.4	145.54	25.02	3.822	117.00	120.82	0.968

Legends

GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

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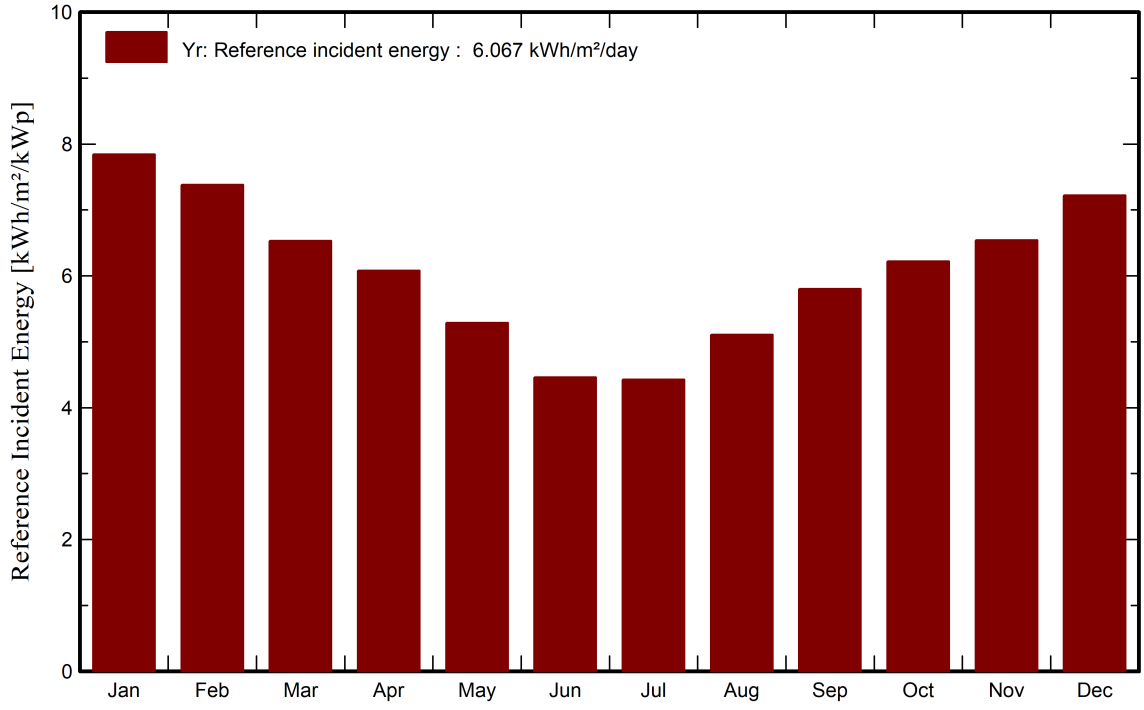
Loss diagram



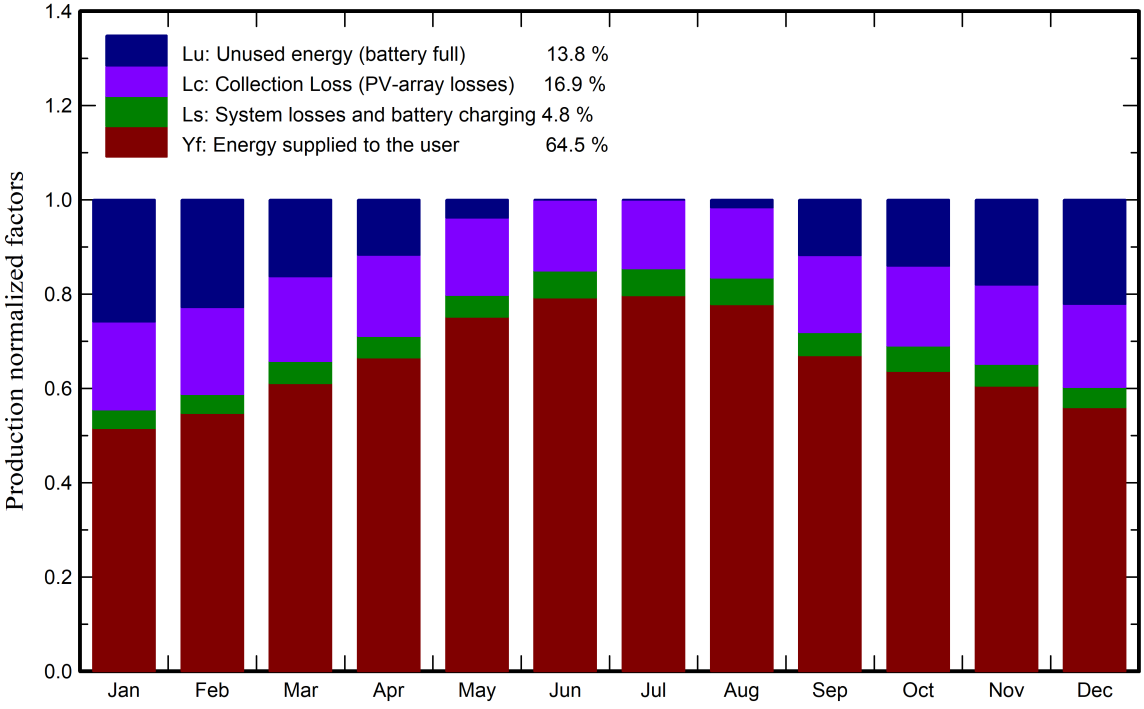
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Predef. graphs

Reference Incident Energy in Collector Plane

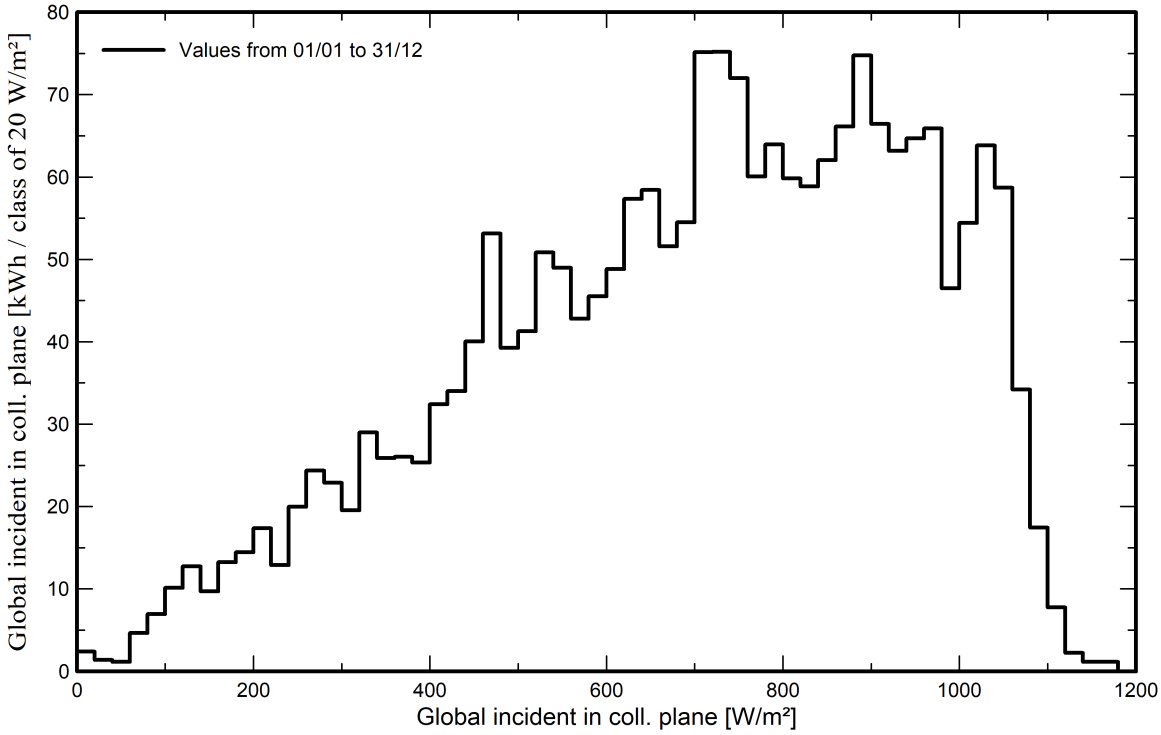


Normalized Production and Loss Factors

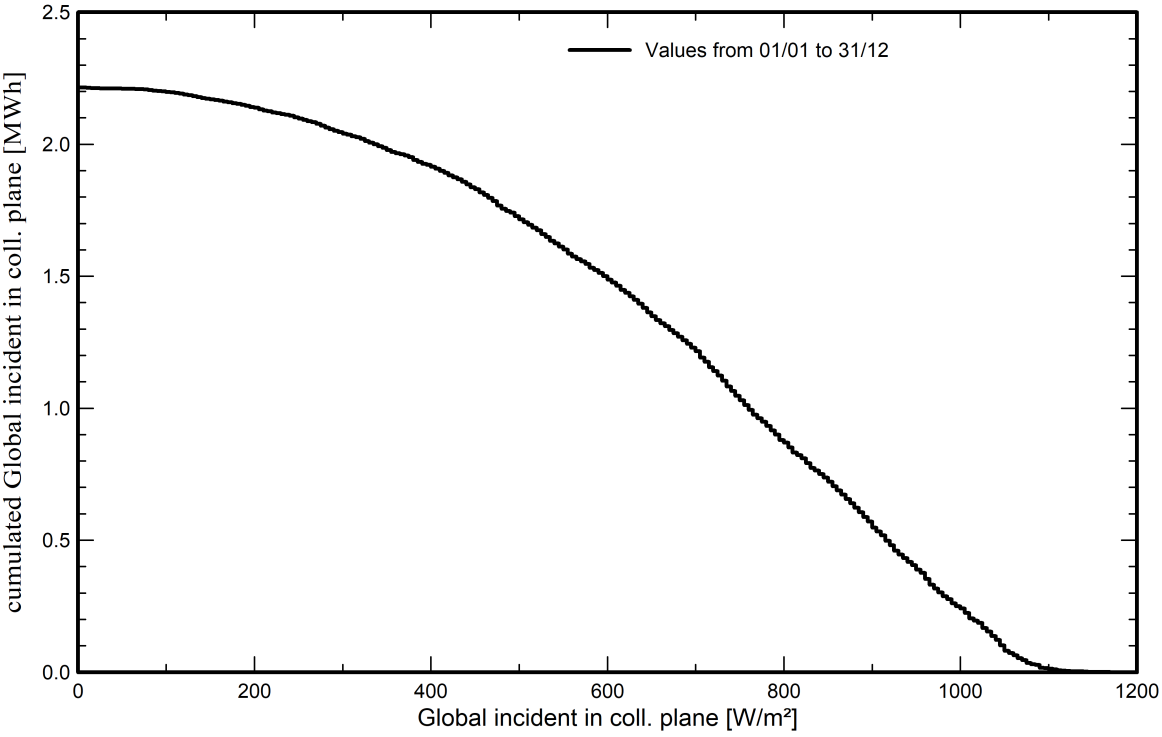


Predef. graphs

Incident Irradiation Distribution

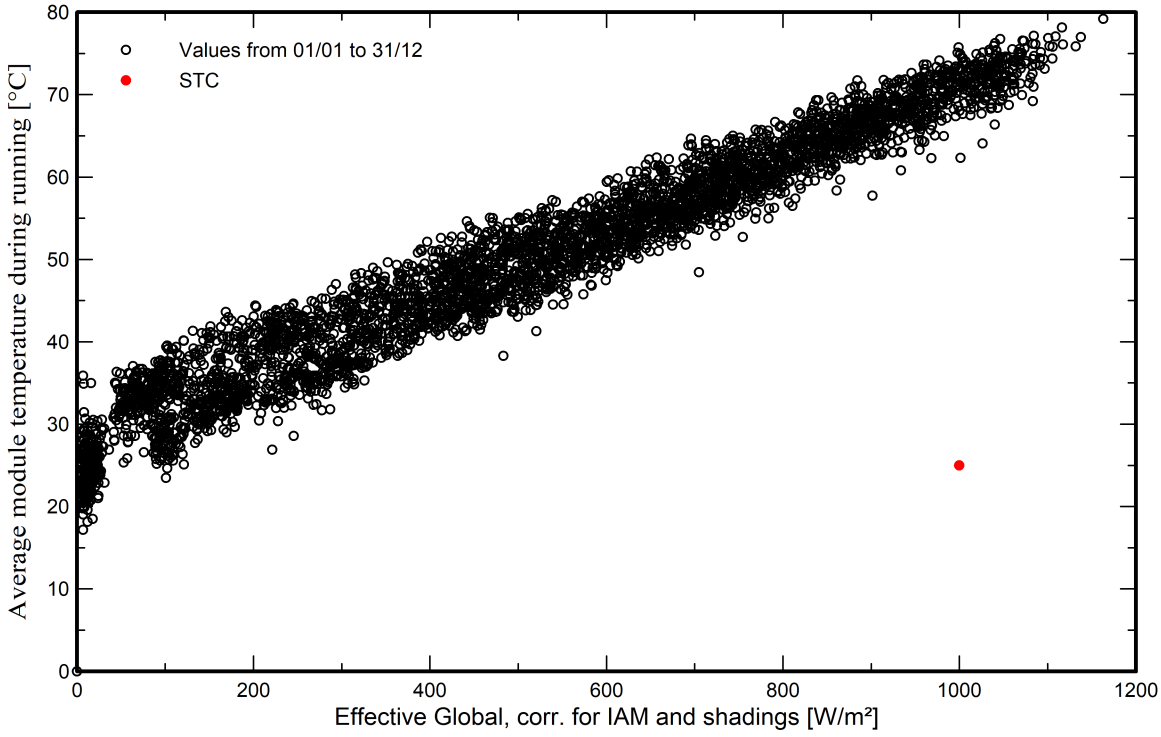


Incident Irradiation cumulative distribution

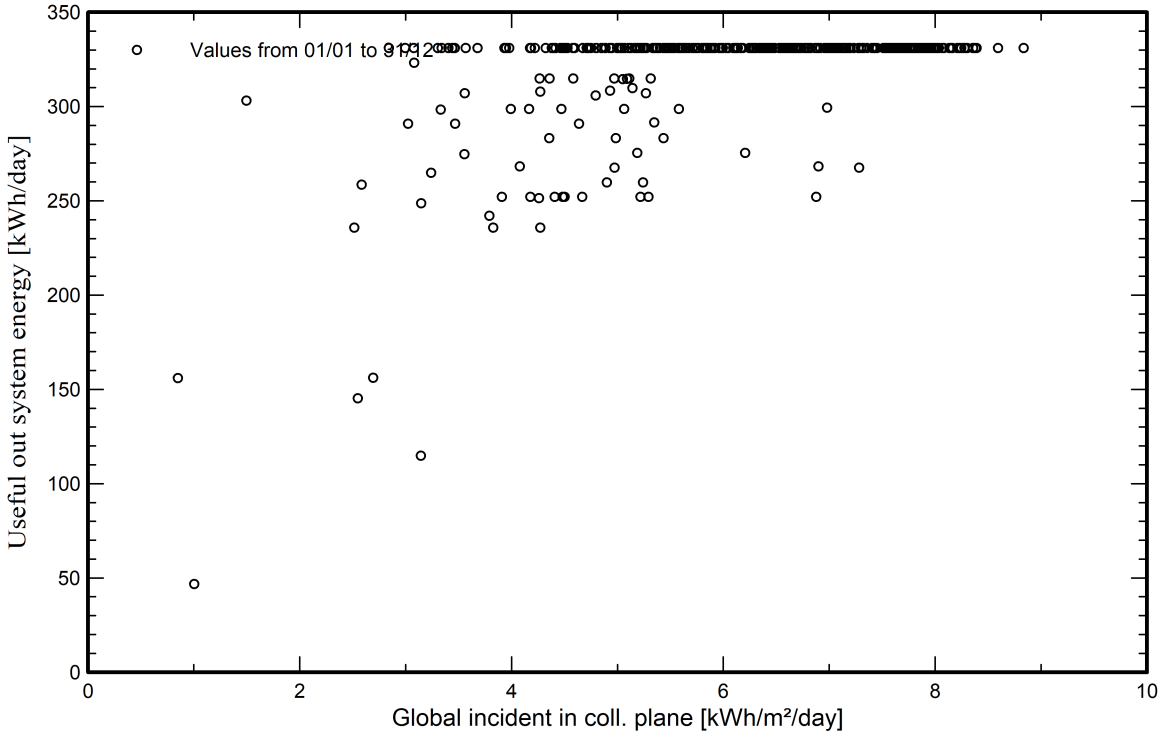


Predef. graphs

Array Temperature vs. Effective Irradiance

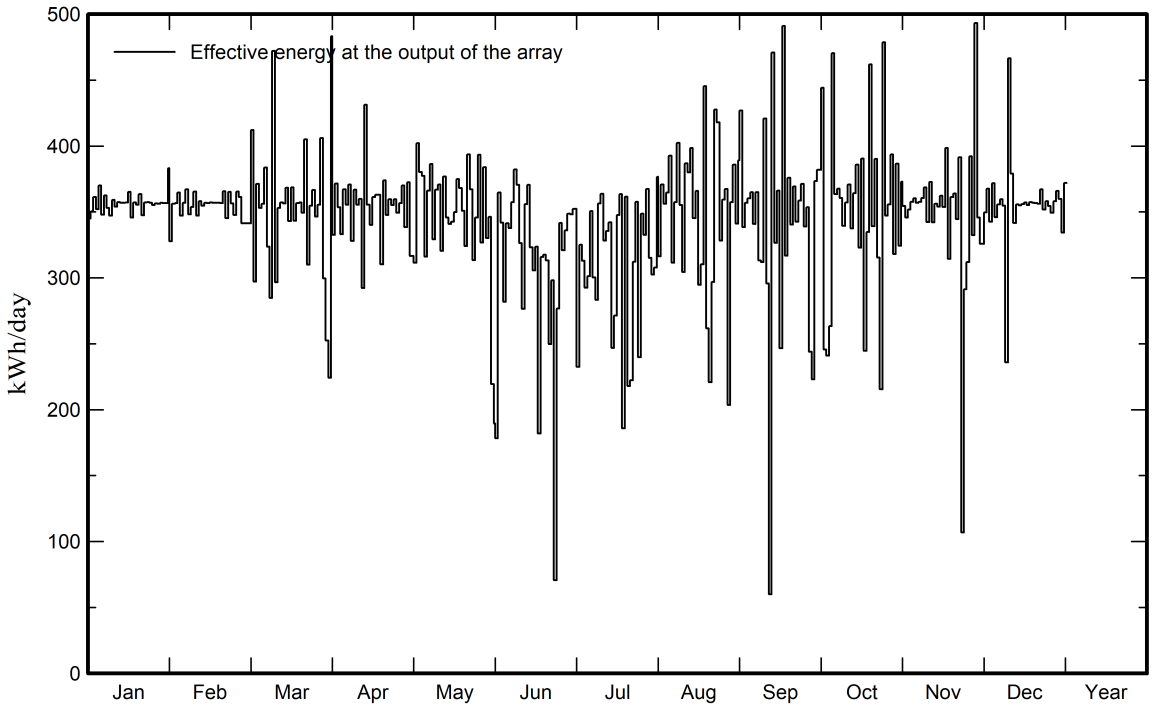


Daily Input/Output diagram

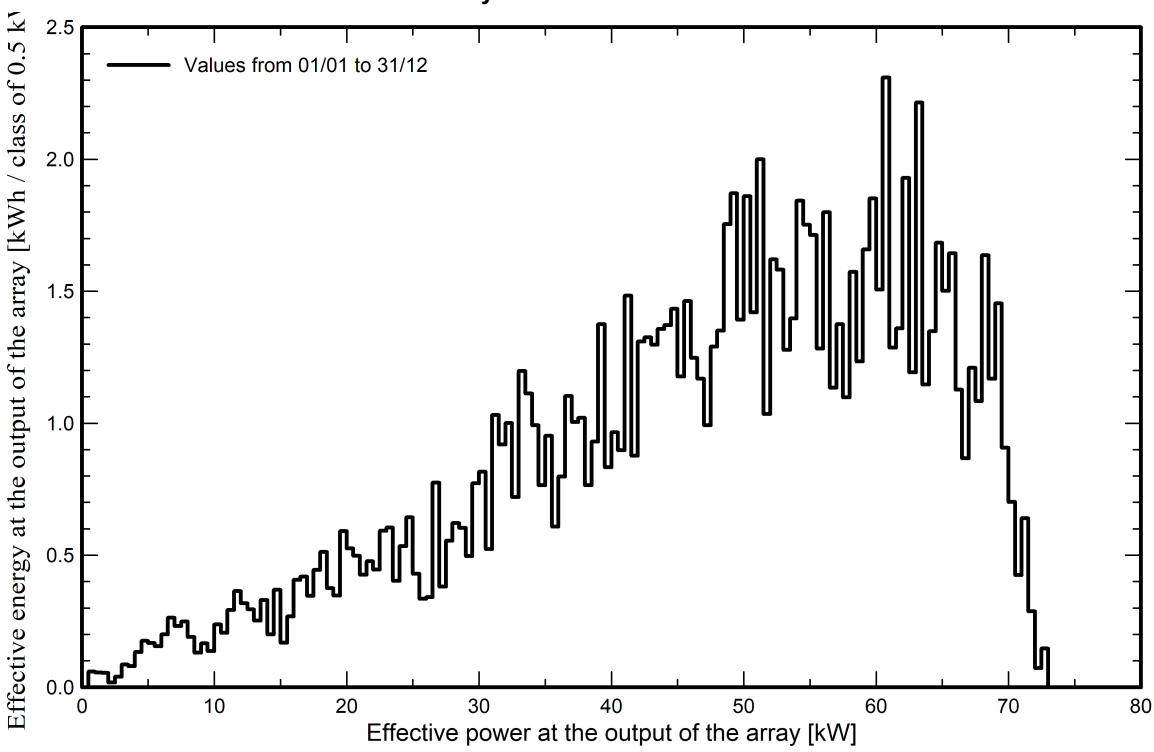


Predef. graphs

Daily Array Output Energy



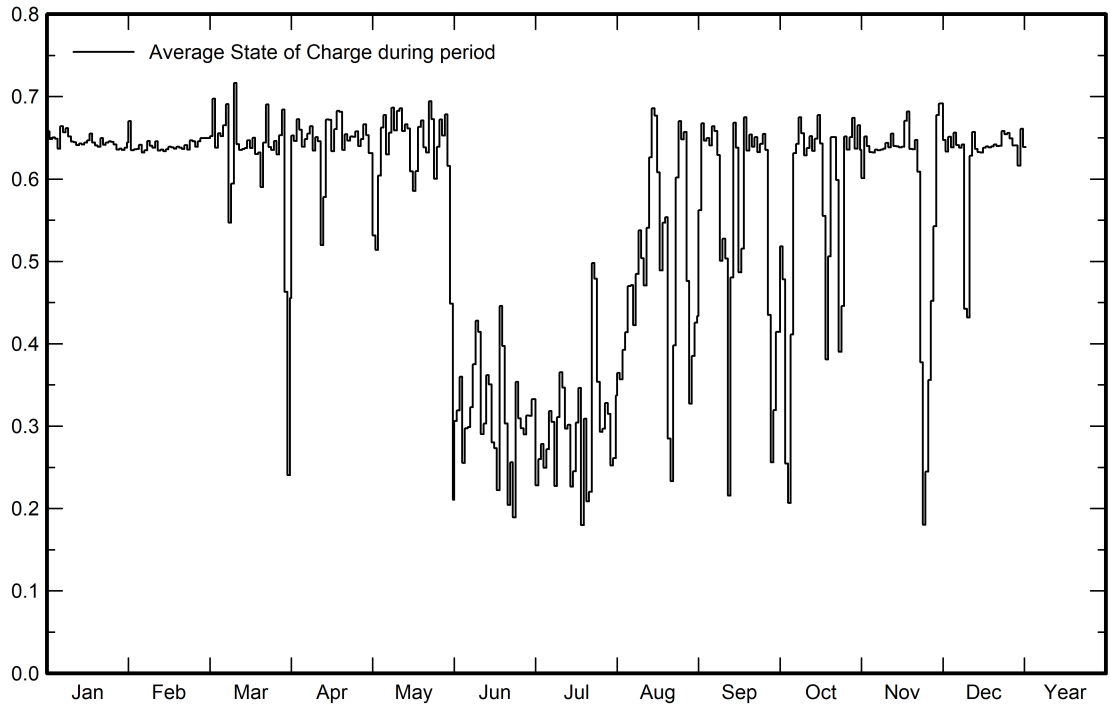
Array Power Distribution



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Predef. graphs

State of charge daily distribution



Cost of the system

Installation costs

Item	Quantity units	Cost Ksh	Total Ksh
PV modules			
JKM585M-7RL4-V	140	11,000.00	1,540,000.00
Supports for modules	140	500.00	70,000.00
Batteries	6	135,000.00	810,000.00
Controllers			35,000.00
Other components			
Accessories, fasteners	2500	1.00	2,500.00
Wiring	40000	1.00	40,000.00
Combiner box	25000	1.00	25,000.00
Monitoring system, display screen	5000	1.00	5,000.00
Measurement system, pyranometer	5000	1.00	5,000.00
Surge arrester	5000	1.00	5,000.00
Installation			
Global installation cost per module	140	500.00	70,000.00
Global installation cost per inverter	1	500.00	500.00
Global installation cost per battery	6	500.00	3,000.00
Settings	1	5,000.00	5,000.00
Taxes			
VAT	1	0.00	25,760.00
		Total	2,641,760.00
		Depreciable asset	2,457,500.00

Operating costs

Item	Total Ksh/year
Total (OPEX)	0.00

System summary

Total installation cost	2,641,760.00	Ksh
Operating costs	0.00	Ksh/year
Excess energy (battery full)	25.0	MWh/year
Used solar energy	117	MWh/year
Used energy cost	0.0000	Ksh/kWh